Compound Statistical Analysis of Block Test Results

Experimental mixes (Exp 1 – Exp 4) versus conventional Concrete Hollow Block (CHB) — source: Stat_Coaching.xlsx

Scope of the data

Five block types (Exp 1, Exp 2, Exp 3, Exp 4 and the CHB control) were each tested with **7 replicate specimens** for eight properties: weight in air, unit weight, compressive strength at 7, 14 and 28 days, screw pull-out, split tensile strength and moisture absorption (5 groups × 7 replicates × 8 tests = 280 observations). All analyses use $\alpha = 0.05$.

Analyses performed

- **Part I** – Descriptive statistics (mean, SD, CV, 95% CI) for every test.
- **Part II** – One-way ANOVA per test, run twice: all five groups (reproduces the workbook) and Exp 1–4 only (isolates differences among the experimental mixes). Includes η^2, ω^2, Levene, Shapiro–Wilk, Welch ANOVA and Kruskal–Wallis checks.
- **Part III** – Tukey HSD / Tukey–Kramer post-hoc comparisons with compact letter groupings.
- **Part IV** – Compound analyses: two-way ANOVA (mix × curing age) for compressive strength, strength-development ratios, MANOVA across all key properties, specimen-level correlation matrix, specific strength, tensile/compressive ratio, performance relative to CHB, and a composite performance index.
- **Part V** – Explanations and interpretation of every result, overall conclusions and limitations.

Key results at a glance

• All Exp mixes are ~67% lighter than CHB (4.66 vs 14.16 kg) yet reach 3.2–4.0× the CHB 28-day compressive strength.

• Every test differs significantly between groups when CHB is included (all $p < 0.0001$); CHB is the inferior group on every mechanical and absorption measure.

• Among the experimental mixes, **Exp 4 is consistently weakest** (compression at all ages, screw pull-out); Exp 1, Exp 2 and Exp 3 are statistically equivalent at 28 days.

• Exp 3 has the lowest moisture absorption; weight, unit weight and split tensile strength do not differ among Exp 1–4.

• Composite index ranking (Exp only): Exp 2 0.85, Exp 3 0.74, Exp 1 0.52, Exp 4 0.10.

Raw data (7 replicates per group)

Test / Group	Replicate values (1 → 7)
Weight in Air (kg)	Exp 1: 4.650, 4.620, 4.660, 4.670, 4.660, 4.690, 4.650
	Exp 2: 4.610, 4.640, 4.670, 4.630, 4.630, 4.680, 4.600
	Exp 3: 4.690, 4.600, 4.630, 4.670, 4.620, 4.660, 4.670
	Exp 4: 4.670, 4.630, 4.690, 4.680, 4.620, 4.650, 4.610
	CHB: 14.100, 13.900, 14.400, 14.200, 14.000, 14.300, 14.200
Unit Weight (kg/m³)	Exp 1: 877.1, 871.5, 879.0, 880.9, 879.0, 884.7, 877.1
	Exp 2: 869.6, 875.2, 880.9, 873.4, 873.4, 882.8, 867.7
	Exp 3: 884.7, 867.7, 873.4, 880.9, 871.5, 879.0, 880.9
	Exp 4: 880.9, 873.4, 884.7, 882.8, 871.5, 877.1, 869.6
	CHB: 1235.0, 1232.0, 1240.0, 1237.0, 1233.0, 1238.0, 1237.0
Compressive 7 d (psi)	Exp 1: 1122, 992, 1084, 1211, 981, 976, 1166
	Exp 2: 1232, 1350, 1007, 1154, 1096, 1222, 1195
	Exp 3: 1214, 1008, 1158, 1092, 1256, 1124, 1136
	Exp 4: 1045, 946, 992, 954, 1125, 985, 968
	CHB: 192, 183, 201, 210, 187, 189, 195
Compressive 14 d (psi)	Exp 1: 1643, 1348, 1591, 1722, 1362, 1355, 1584
	Exp 2: 1708, 1688, 1455, 1385, 1352, 1421, 1395
	Exp 3: 1504, 1225, 1436, 1285, 1432, 1354, 1348
	Exp 4: 1235, 1102, 1284, 1087, 1452, 1188, 1122
	CHB: 381, 395, 370, 385, 387, 377, 365
Compressive 28 d (psi)	Exp 1: 1972, 1624, 1827, 2030, 1668, 1653, 1871
	Exp 2: 1929, 1943, 1668, 1711, 1697, 1832, 1786
	Exp 3: 1842, 1595, 1769, 1581, 1711, 1668, 1697
	Exp 4: 1581, 1363, 1523, 1349, 1668, 1378, 1392
	CHB: 485, 492, 420, 434, 472, 445, 452
Screw Pull-out (kg)	Exp 1: 28.0, 24.0, 27.0, 29.0, 24.0, 24.0, 27.0
	Exp 2: 32.0, 31.0, 25.0, 26.0, 25.0, 28.0, 27.0
	Exp 3: 29.0, 24.0, 26.0, 23.0, 28.0, 27.0, 27.0
	Exp 4: 24.0, 22.0, 25.0, 23.0, 26.0, 24.0, 22.0

Test / Group	Replicate values (1 → 7)
	CHB: 11.0, 9.0, 10.0, 12.0, 7.0, 8.0, 9.0
Split Tensile (MPa)	Exp 1: 0.450, 0.320, 0.430, 0.470, 0.330, 0.330, 0.420
	Exp 2: 0.460, 0.440, 0.330, 0.340, 0.330, 0.360, 0.350
	Exp 3: 0.430, 0.370, 0.420, 0.380, 0.410, 0.360, 0.360
	Exp 4: 0.370, 0.320, 0.360, 0.310, 0.380, 0.330, 0.320
	CHB: 0.110, 0.090, 0.130, 0.100, 0.080, 0.140, 0.120
Moisture Absorption (%)	Exp 1: 8.50, 9.10, 8.70, 8.20, 8.80, 8.90, 8.40
	Exp 2: 8.10, 8.20, 8.90, 8.30, 8.10, 8.50, 8.40
	Exp 3: 8.30, 8.10, 8.00, 8.40, 8.40, 8.10, 8.20
	Exp 4: 8.60, 8.50, 8.70, 8.80, 8.40, 8.30, 8.90
	CHB: 13.10, 12.70, 14.20, 13.50, 12.90, 13.70, 13.20

Part I — Descriptive Statistics

Test	Group	n	Mean	SD	CV %	Min	Max	95% CI of mean
Weight in Air (kg)	Exp 1	7	4.657	0.021	0.5	4.620	4.690	4.637 – 4.677
	Exp 2	7	4.637	0.029	0.6	4.600	4.680	4.610 – 4.664
	Exp 3	7	4.649	0.032	0.7	4.600	4.690	4.619 – 4.679
	Exp 4	7	4.650	0.031	0.7	4.610	4.690	4.621 – 4.679
	CHB	7	14.157	0.172	1.2	13.900	14.400	13.998 – 14.316
Unit Weight (kg/m³)	Exp 1	7	878.5	4.0	0.5	871.5	884.7	874.7 – 882.2
	Exp 2	7	874.7	5.5	0.6	867.7	882.8	869.6 – 879.8
	Exp 3	7	876.9	6.1	0.7	867.7	884.7	871.2 – 882.5
	Exp 4	7	877.1	5.9	0.7	869.6	884.7	871.7 – 882.5
	CHB	7	1236.0	2.8	0.2	1232.0	1240.0	1233.4 – 1238.6
Compressive 7 d (psi)	Exp 1	7	1076	95.4	8.9	976	1211	988 – 1164
	Exp 2	7	1179	109.0	9.2	1007	1350	1079 – 1280
	Exp 3	7	1141	80.9	7.1	1008	1256	1066 – 1216
	Exp 4	7	1002	63.2	6.3	946	1125	944 – 1061
	CHB	7	194	9.2	4.7	183	210	185 – 202
Compressive 14 d (psi)	Exp 1	7	1515	156.3	10.3	1348	1722	1370 – 1660
	Exp 2	7	1486	148.2	10.0	1352	1708	1349 – 1623
	Exp 3	7	1369	95.9	7.0	1225	1504	1280 – 1458
	Exp 4	7	1210	128.9	10.7	1087	1452	1091 – 1329
	CHB	7	380	10.3	2.7	365	395	370 – 390
Compressive 28 d (psi)	Exp 1	7	1806	162.3	9.0	1624	2030	1656 – 1956
	Exp 2	7	1795	111.1	6.2	1668	1943	1692 – 1898
	Exp 3	7	1695	92.3	5.4	1581	1842	1609 – 1780
	Exp 4	7	1465	125.7	8.6	1349	1668	1349 – 1581
	CHB	7	457	26.8	5.9	420	492	432 – 482
Screw Pull-out (kg)	Exp 1	7	26.1	2.1	8.1	24.0	29.0	24.2 – 28.1
	Exp 2	7	27.7	2.8	10.1	25.0	32.0	25.1 – 30.3
	Exp 3	7	26.3	2.1	8.1	23.0	29.0	24.3 – 28.3
	Exp 4	7	23.7	1.5	6.3	22.0	26.0	22.3 – 25.1
	CHB	7	9.4	1.7	18.2	7.0	12.0	7.8 – 11.0
Split Tensile (MPa)	Exp 1	7	0.393	0.064	16.3	0.320	0.470	0.334 – 0.452
	Exp 2	7	0.373	0.054	14.5	0.330	0.460	0.323 – 0.423
	Exp 3	7	0.390	0.029	7.5	0.360	0.430	0.363 – 0.417
	Exp 4	7	0.341	0.028	8.2	0.310	0.380	0.316 – 0.367
	CHB	7	0.110	0.022	19.6	0.080	0.140	0.090 – 0.130
Moisture Absorption (%)	Exp 1	7	8.66	0.31	3.6	8.20	9.10	8.37 – 8.94
	Exp 2	7	8.36	0.28	3.4	8.10	8.90	8.10 – 8.62
	Exp 3	7	8.21	0.16	1.9	8.00	8.40	8.07 – 8.36
	Exp 4	7	8.60	0.22	2.5	8.30	8.90	8.40 – 8.80
	CHB	7	13.33	0.51	3.8	12.70	14.20	12.85 – 13.80

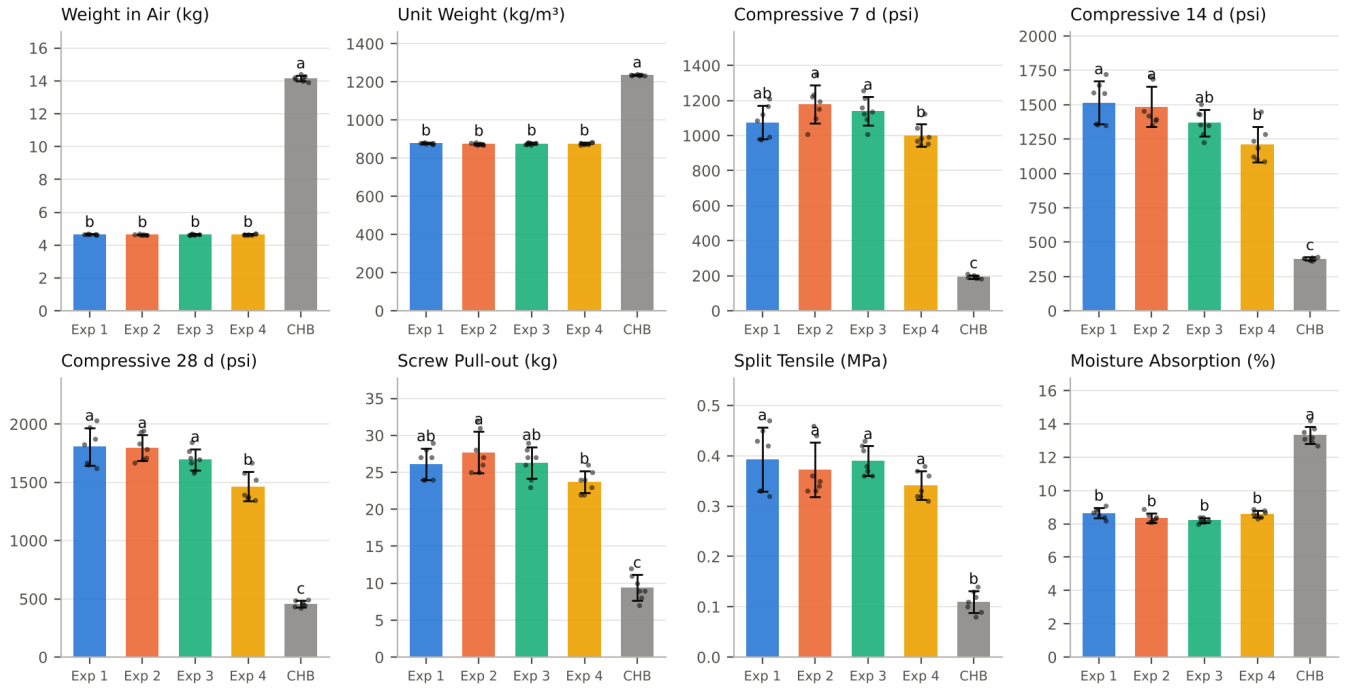


Figure 1. Group means (bars) ± 1 SD with individual specimens (dots). Letters are Tukey HSD groupings across all five groups: groups sharing a letter are not significantly different ($\alpha = 0.05$).

Part II — One-way ANOVA

A. All five groups (Exp 1–4 and CHB)

Test	F	df	p-value	η²	ω²	Levene p	Shapiro p	Welch p	Decision
Weight in Air (kg)	19266.18	4, 30	< 0.0001	1.000	1.000	0.0012	< 0.0001	< 0.0001	Reject H0
Unit Weight (kg/m³)	7144.91	4, 30	< 0.0001	0.999	0.999	0.3587	0.5195	< 0.0001	Reject H0
Compressive 7 d (psi)	186.83	4, 30	< 0.0001	0.961	0.955	0.0558	0.7853	< 0.0001	Reject H0
Compressive 14 d (psi)	106.73	4, 30	< 0.0001	0.934	0.924	0.1187	0.0979	< 0.0001	Reject H0
Compressive 28 d (psi)	177.44	4, 30	< 0.0001	0.959	0.953	0.0642	0.3749	< 0.0001	Reject H0
Screw Pull-out (kg)	89.70	4, 30	< 0.0001	0.923	0.910	0.6856	0.3428	< 0.0001	Reject H0
Split Tensile (MPa)	55.16	4, 30	< 0.0001	0.880	0.861	0.1890	0.4015	< 0.0001	Reject H0
Moisture Absorption (%)	328.24	4, 30	< 0.0001	0.978	0.974	0.1608	0.6113	< 0.0001	Reject H0

η² = eta squared, ω² = omega squared (effect sizes). Levene (median-centred) tests equal variances; Shapiro–Wilk tests normality of residuals; Welch ANOVA is robust to unequal variances.

Full ANOVA source tables (all groups)

Test	Source	SS	df	MS	F	p-value
Weight in Air (kg)	Between groups	506.4	4	126.6	19266.18	< 0.0001
	Within groups	0.1971	30	0.0066		
	Total	506.5	34			
Unit Weight (kg/m³)	Between groups	722,652.4	4	180,663.1	7144.91	< 0.0001
	Within groups	758.6	30	25.3		
	Total	723,411.0	34			
Compressive 7 d (psi)	Between groups	4,721,942.5	4	1,180,485.6	186.83	< 0.0001
	Within groups	189,552.3	30	6,318.4		
	Total	4,911,494.7	34			
Compressive 14 d (psi)	Between groups	6,173,864.5	4	1,543,466.1	106.73	< 0.0001
	Within groups	433,840.3	30	14,461.3		
	Total	6,607,704.7	34			
Compressive 28 d (psi)	Between groups	9,042,844.2	4	2,260,711.0	177.44	< 0.0001
	Within groups	382,229.7	30	12,741.0		
	Total	9,425,073.9	34			
Screw Pull-out (kg)	Between groups	1,589.0	4	397.3	89.70	< 0.0001
	Within groups	132.9	30	4.4286		
	Total	1,721.9	34			
Split Tensile (MPa)	Between groups	0.4029	4	0.1007	55.16	< 0.0001
	Within groups	0.0548	30	0.0018		
	Total	0.4576	34			
Moisture Absorption (%)	Between groups	133.8	4	33.4	328.24	< 0.0001
	Within groups	3.0571	30	0.1019		
	Total	136.9	34			

B. Experimental mixes only (Exp 1–4)

Because CHB is so different from the experimental blocks, it dominates the all-group ANOVA. Removing it tests the more useful question: *do the four experimental mixes differ from each other?*

Test	F	df	p-value	η²	ω²	Levene p	Shapiro p	Welch p	K–W p	Decision
Weight in Air (kg)	0.58	3, 24	0.6360	0.067	0.000	0.5846	0.2880	0.5974	0.7185	Not significant
Unit Weight (kg/m³)	0.58	3, 24	0.6359	0.067	0.000	0.5853	0.2888	0.5973	0.7185	Not significant
Compressive 7 d (psi)	5.38	3, 24	0.0056	0.402	0.319	0.5443	0.9944	0.0076	0.0085	Significant
Compressive 14 d (psi)	7.45	3, 24	0.0011	0.482	0.409	0.7616	0.0505	0.0068	0.0072	Significant
Compressive 28 d (psi)	11.16	3, 24	< 0.0001	0.582	0.521	0.4773	0.1283	0.0012	0.0023	Significant
Screw Pull-out (kg)	4.02	3, 24	0.0189	0.334	0.244	0.6308	0.3282	0.0192	0.0294	Significant
Split Tensile (MPa)	1.80	3, 24	0.1735	0.184	0.079	0.3016	0.2616	0.0500	0.1658	Not significant
Moisture Absorption (%)	4.89	3, 24	0.0086	0.379	0.294	0.5230	0.8647	0.0075	0.0127	Significant

K–W = Kruskal–Wallis (non-parametric check). Negative ω² values are reported as 0.000.

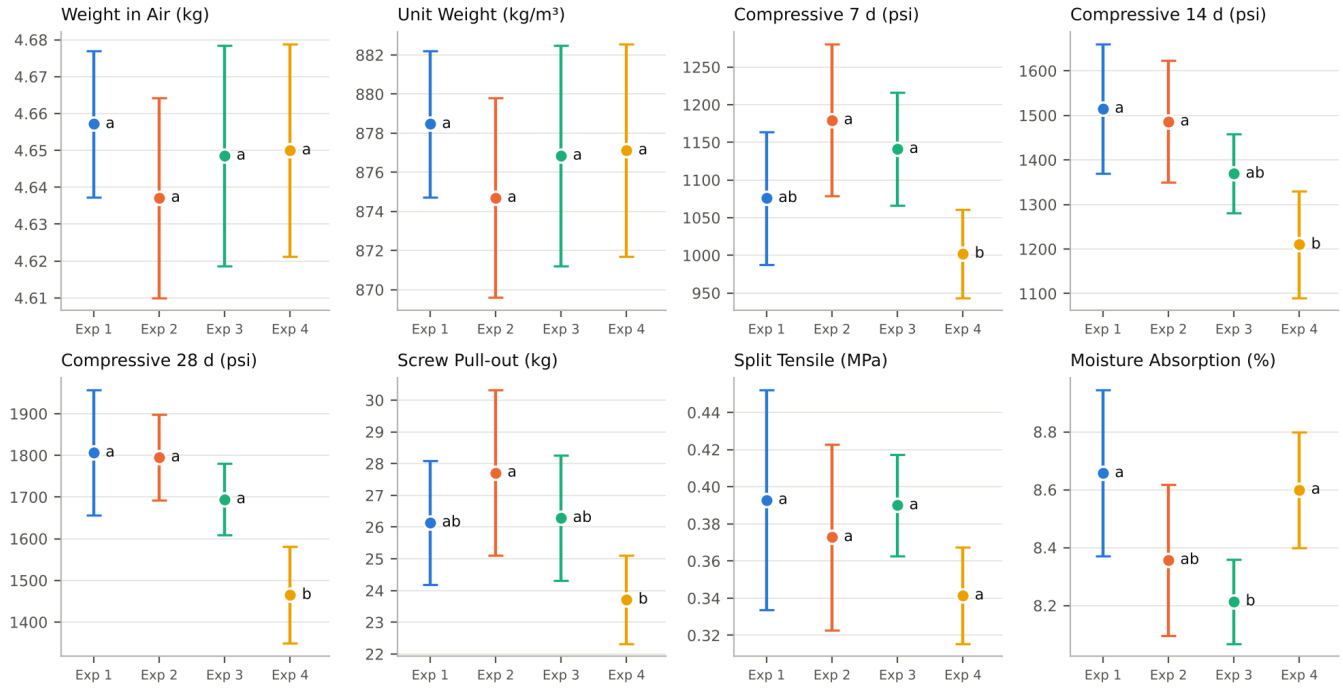


Figure 2. Experimental mixes only: means with 95% confidence intervals. Letters are Tukey HSD groupings among Exp 1–4.

Part III — Tukey HSD Post-hoc Comparisons

A. Compact letter groupings

Test	Exp 1	Exp 2	Exp 3	Exp 4	CHB	Exp-only groups
Weight in Air (kg)	4.657 b	4.637 b	4.649 b	4.650 b	14.157 a	1:a 2:a 3:a 4:a
Unit Weight (kg/m³)	878.5 b	874.7 b	876.9 b	877.1 b	1236.0 a	1:a 2:a 3:a 4:a
Compressive 7 d (psi)	1076 ab	1179 a	1141 a	1002 b	194 c	1:ab 2:a 3:a 4:b
Compressive 14 d (psi)	1515 a	1486 a	1369 ab	1210 b	380 c	1:a 2:a 3:ab 4:b
Compressive 28 d (psi)	1806 a	1795 a	1695 a	1465 b	457 c	1:a 2:a 3:a 4:b
Screw Pull-out (kg)	26.1 ab	27.7 a	26.3 ab	23.7 b	9.4 c	1:ab 2:a 3:ab 4:b
Split Tensile (MPa)	0.393 a	0.373 a	0.390 a	0.341 a	0.110 b	1:a 2:a 3:a 4:a
Moisture Absorption (%)	8.66 b	8.36 b	8.21 b	8.60 b	13.33 a	1:a 2:ab 3:b 4:a

Values are means followed by the all-group Tukey letter. "Exp-only groups" gives letters from the Exp 1–4 analysis (1 = Exp 1 … 4 = Exp 4).

B. Pairwise comparisons among experimental mixes (Exp 1–4)

Test	Comparison	Mean diff.	95% CI	p (Tukey)	Cohen's d	Result
Weight in Air (kg)	Exp 1 – Exp 2	0.020	-0.023 to 0.063	0.5736	0.69	n.s.
	Exp 1 – Exp 3	0.009	-0.034 to 0.051	0.9440	0.30	n.s.
	Exp 1 – Exp 4	0.007	-0.035 to 0.050	0.9663	0.25	n.s.
	Exp 2 – Exp 3	-0.011	-0.054 to 0.031	0.8795	-0.40	n.s.
	Exp 2 – Exp 4	-0.013	-0.055 to 0.030	0.8379	-0.45	n.s.
	Exp 3 – Exp 4	-0.001	-0.044 to 0.041	0.9997	-0.05	n.s.
Unit Weight (kg/m³)	Exp 1 – Exp 2	3.8	-4.3 to 11.8	0.5735	0.69	n.s.
	Exp 1 – Exp 3	1.6	-6.4 to 9.6	0.9440	0.30	n.s.
	Exp 1 – Exp 4	1.3	-6.7 to 9.4	0.9663	0.25	n.s.
	Exp 2 – Exp 3	-2.2	-10.2 to 5.9	0.8795	-0.40	n.s.
	Exp 2 – Exp 4	-2.4	-10.4 to 5.6	0.8378	-0.45	n.s.
	Exp 3 – Exp 4	-0.3	-8.3 to 7.8	0.9997	-0.05	n.s.
Compressive 7 d (psi)	Exp 1 – Exp 2	-103	-234 to 27.4	0.1575	-1.17	n.s.
	Exp 1 – Exp 3	-65.1	-196 to 65.7	0.5275	-0.73	n.s.
	Exp 1 – Exp 4	73.9	-57.0 to 205	0.4211	0.83	n.s.
	Exp 2 – Exp 3	38.3	-92.6 to 169	0.8505	0.43	n.s.
	Exp 2 – Exp 4	177	46.4 to 308	0.0053	2.00	Significant
	Exp 3 – Exp 4	139	8.1 to 270	0.0345	1.57	Significant
Compressive 14 d (psi)	Exp 1 – Exp 2	28.7	-169 to 227	0.9779	0.21	n.s.
	Exp 1 – Exp 3	146	-52.2 to 344	0.2048	1.09	n.s.
	Exp 1 – Exp 4	305	107 to 503	0.0015	2.27	Significant
	Exp 2 – Exp 3	117	-81.0 to 315	0.3809	0.87	n.s.
	Exp 2 – Exp 4	276	78.2 to 474	0.0040	2.06	Significant
	Exp 3 – Exp 4	159	-39.0 to 357	0.1475	1.18	n.s.
Compressive 28 d (psi)	Exp 1 – Exp 2	11.3	-174 to 196	0.9983	0.09	n.s.
	Exp 1 – Exp 3	112	-73.3 to 297	0.3631	0.89	n.s.
	Exp 1 – Exp 4	342	157 to 527	0.0002	2.72	Significant
	Exp 2 – Exp 3	100	-84.6 to 285	0.4546	0.80	n.s.
	Exp 2 – Exp 4	330	145 to 515	0.0003	2.63	Significant
	Exp 3 – Exp 4	230	44.8 to 415	0.0111	1.83	Significant
Screw Pull-out (kg)	Exp 1 – Exp 2	-1.6	-4.8 to 1.7	0.5462	-0.72	n.s.
	Exp 1 – Exp 3	-0.1	-3.4 to 3.1	0.9993	-0.07	n.s.
	Exp 1 – Exp 4	2.4	-0.8 to 5.7	0.1901	1.11	n.s.
	Exp 2 – Exp 3	1.4	-1.8 to 4.7	0.6205	0.65	n.s.
	Exp 2 – Exp 4	4.0	0.8 to 7.2	0.0113	1.83	Significant
	Exp 3 – Exp 4	2.6	-0.7 to 5.8	0.1529	1.17	n.s.
Split Tensile (MPa)	Exp 1 – Exp 2	0.020	-0.049 to 0.089	0.8519	0.43	n.s.

Test	Comparison	Mean diff.	95% CI	p (Tukey)	Cohen's d	Result
	Exp 1 – Exp 3	0.003	-0.066 to 0.071	0.9994	0.06	n.s.
	Exp 1 – Exp 4	0.051	-0.017 to 0.120	0.1923	1.11	n.s.
	Exp 2 – Exp 3	-0.017	-0.086 to 0.051	0.9002	-0.37	n.s.
	Exp 2 – Exp 4	0.031	-0.037 to 0.100	0.5940	0.68	n.s.
	Exp 3 – Exp 4	0.049	-0.020 to 0.117	0.2336	1.04	n.s.
Moisture Absorption (%)	Exp 1 – Exp 2	0.30	-0.07 to 0.67	0.1365	1.21	n.s.
	Exp 1 – Exp 3	0.44	0.08 to 0.81	0.0138	1.78	Significant
	Exp 1 – Exp 4	0.06	-0.31 to 0.42	0.9727	0.23	n.s.
	Exp 2 – Exp 3	0.14	-0.22 to 0.51	0.7076	0.57	n.s.
	Exp 2 – Exp 4	-0.24	-0.61 to 0.12	0.2853	-0.98	n.s.
	Exp 3 – Exp 4	-0.39	-0.75 to -0.02	0.0366	-1.55	Significant

Shaded rows are significant at $\alpha = 0.05$. Cohen's d uses the pooled within-group SD ($\sqrt{MSW}$); $|d| \geq 0.8$ is conventionally "large".

C. Each experimental mix versus CHB (all-group Tukey)

Test	Exp 1 vs CHB	Exp 2 vs CHB	Exp 3 vs CHB	Exp 4 vs CHB
Weight in Air (kg)	-67.1% (p < 0.0001)	-67.2% (p < 0.0001)	-67.2% (p < 0.0001)	-67.2% (p < 0.0001)
Unit Weight (kg/m³)	-28.9% (p < 0.0001)	-29.2% (p < 0.0001)	-29.1% (p < 0.0001)	-29.0% (p < 0.0001)
Compressive 7 d (psi)	+455.0% (p < 0.0001)	+508.4% (p < 0.0001)	+488.7% (p < 0.0001)	+416.9% (p < 0.0001)
Compressive 14 d (psi)	+298.7% (p < 0.0001)	+291.1% (p < 0.0001)	+260.3% (p < 0.0001)	+218.4% (p < 0.0001)
Compressive 28 d (psi)	+295.2% (p < 0.0001)	+292.7% (p < 0.0001)	+270.7% (p < 0.0001)	+220.4% (p < 0.0001)
Screw Pull-out (kg)	+177.3% (p < 0.0001)	+193.9% (p < 0.0001)	+178.8% (p < 0.0001)	+151.5% (p < 0.0001)
Split Tensile (MPa)	+257.1% (p < 0.0001)	+239.0% (p < 0.0001)	+254.5% (p < 0.0001)	+210.4% (p < 0.0001)
Moisture Absorption (%)	-35.0% (p < 0.0001)	-37.3% (p < 0.0001)	-38.4% (p < 0.0001)	-35.5% (p < 0.0001)

Percentage = (Exp mean – CHB mean) / CHB mean. All 32 Exp-vs-CHB comparisons are significant.

Part IV — Compound Analyses

A. Two-way ANOVA: mix × curing age (compressive strength)

All five groups (n = 105)

Source	SS	df	F	p-value	Partial η^2
Mix	19,455,847	4	435.31	< 0.0001	0.951
Curing age	4,828,886	2	216.08	< 0.0001	0.828
Mix × Age	482,804	8	5.40	< 0.0001	0.324
Residual	1,005,622	90			

Exp 1–4 only (n = 84)

Source	SS	df	F	p-value	Partial η^2
Mix	885,936	3	21.26	< 0.0001	0.470
Curing age	4,883,435	2	175.77	< 0.0001	0.830
Mix × Age	171,775	6	2.06	0.0685	0.147
Residual	1,000,183	72			

B. Strength development

Group	7 d (psi)	14 d (psi)	28 d (psi)	7 d as % of 28 d	14 d as % of 28 d	Gain 7→28 d
Exp 1	1076	1515	1806	59.6%	83.9%	+67.9%
Exp 2	1179	1486	1795	65.7%	82.8%	+52.2%
Exp 3	1141	1369	1695	67.3%	80.8%	+48.5%
Exp 4	1002	1210	1465	68.4%	82.6%	+46.2%
CHB	194	380	457	42.4%	83.1%	+135.8%

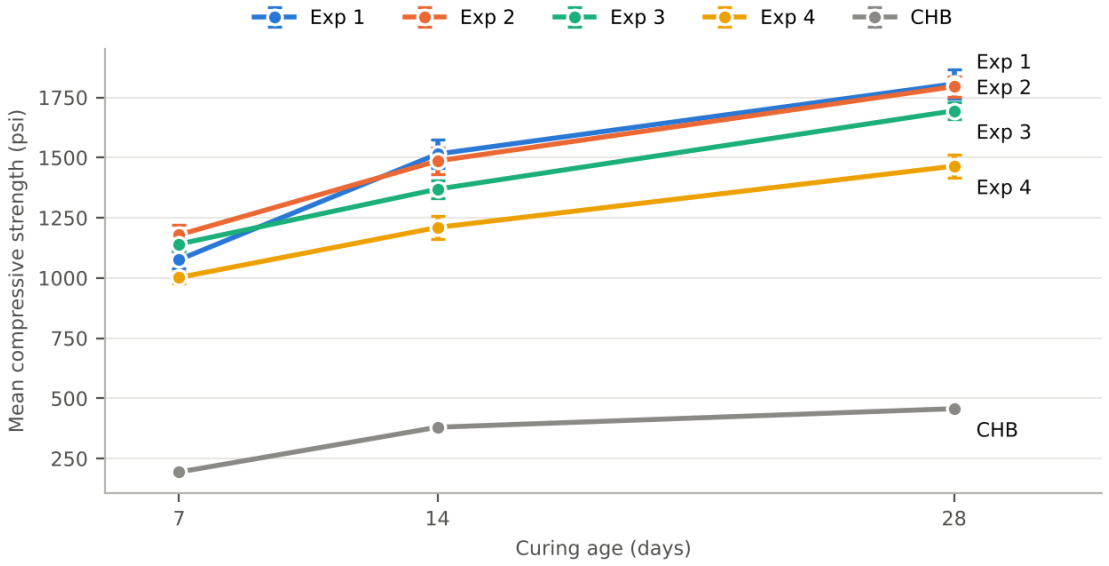


Figure 3. Mean compressive strength by curing age (± 1 standard error).

C. Multivariate ANOVA (MANOVA)

Dependent variables tested jointly: 28-day compressive strength, split tensile strength, screw pull-out, moisture absorption and unit weight.

Statistic	Exp 1–4: value	F	df	p-value	All groups: value	F	p-value
Wilks' lambda	0.0556	6.85	15, 55.6	< 0.0001	9.713e-05	66.31	< 0.0001
Pillai's trace	1.6449	5.34	15, 66.0	< 0.0001	2.33	8.09	< 0.0001
Hotelling-Lawley trace	6.5175	8.33	15, 32.9	< 0.0001	1083	1353.00	< 0.0001
Roy's greatest root	4.9084	21.60	5, 22.0	< 0.0001	1078	6250.63	< 0.0001

D. Specimen-level correlations (Exp 1–4, n = 28)

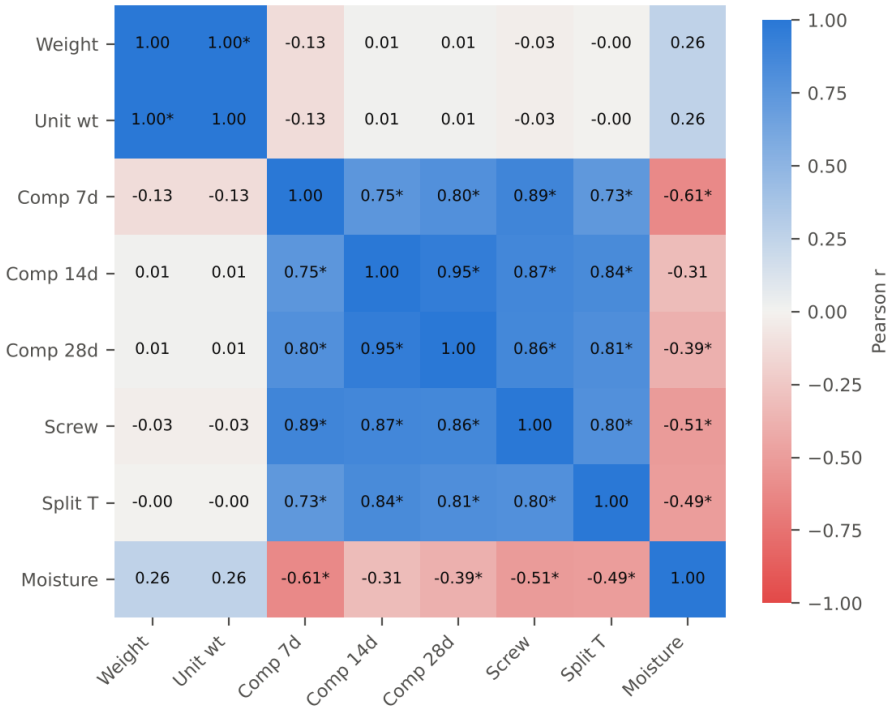


Figure 4. Pearson correlation matrix across the 28 experimental specimens. * = significant at α = 0.05. Replicate i of each test is treated as the same specimen, as laid out in the "Raw Files" sheet.

E. Efficiency ratios

Group	Specific strength (psi per kg/m³)	Split tensile / compressive (28 d)	Weight reduction vs CHB	Strength gain vs CHB (28 d)
Exp 1	2.056 ± 0.184	3.14%	67.1%	3.95×
Exp 2	2.053 ± 0.130	3.00%	67.2%	3.93×
Exp 3	1.933 ± 0.101	3.34%	67.2%	3.71×
Exp 4	1.670 ± 0.144	3.38%	67.2%	3.20×
CHB	0.370 ± 0.022	3.52%	—	—

F. Performance relative to CHB

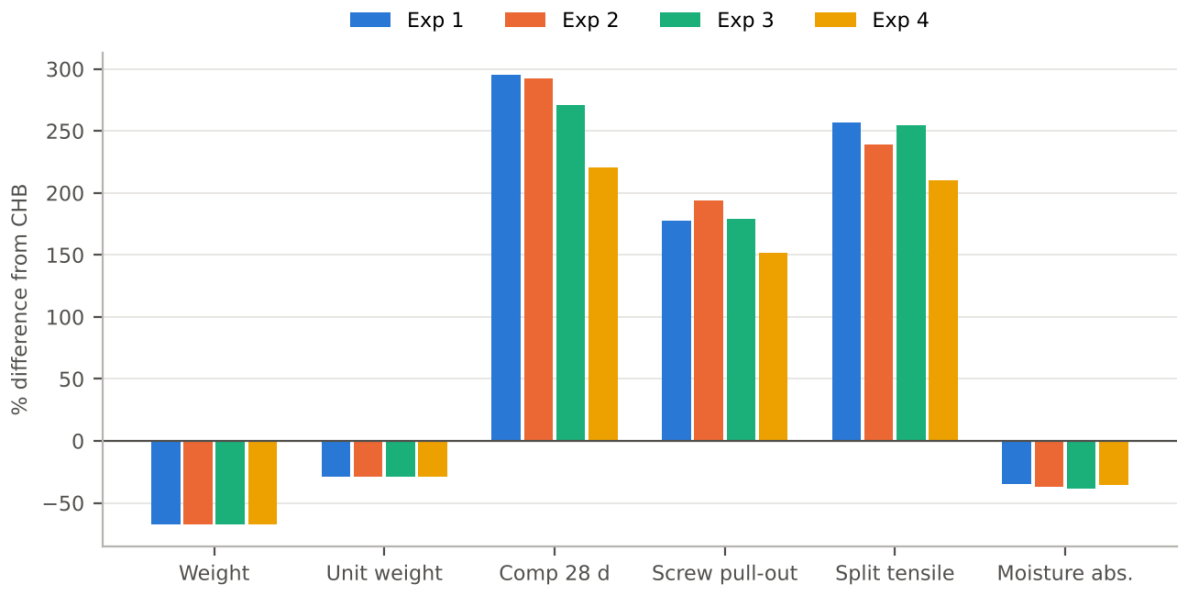


Figure 5. Percentage difference of each experimental mix from the CHB mean. Negative values are favourable for weight, unit weight and moisture absorption; positive values are favourable for the strength properties.

G. Composite performance index (Exp 1–4)

Group	Comp 28 d	Split tensile	Screw pull-out	Low absorption	Low unit weight	Index	Rank
Exp 1	1.00	1.00	0.61	0.00	0.00	0.521	3
Exp 2	0.97	0.61	1.00	0.68	1.00	0.851	1
Exp 3	0.67	0.94	0.64	1.00	0.43	0.738	2
Exp 4	0.00	0.00	0.00	0.13	0.36	0.097	4

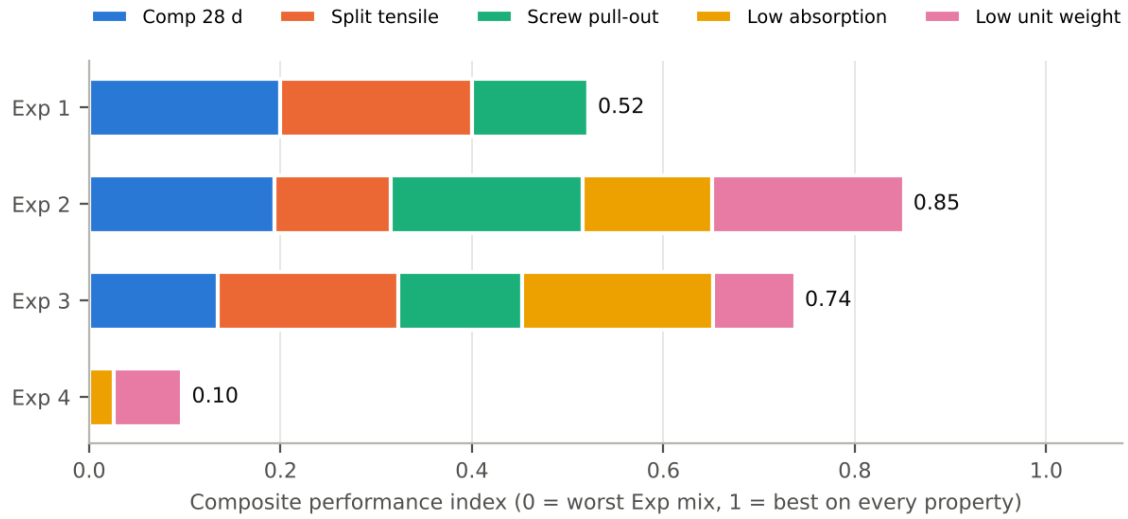


Figure 6. Composite index = equal-weighted mean of min–max scores (best Exp mix = 1, worst = 0) for each property; segment widths show each property's contribution.

Part V — Explanations and Interpretation

1. What each statistical tool means

Mean, SD and CV. The mean is the average of the 7 replicates. The standard deviation (SD) shows how spread out the replicates are. The coefficient of variation (CV = SD ÷ mean × 100) expresses that spread as a percentage, which allows consistency to be compared between tests with different units. CV below ~10% indicates good production/testing consistency.

95% confidence interval. The range that, with 95% confidence, contains the true mean of the block type. Non-overlapping intervals usually indicate a real difference.

One-way ANOVA. Tests H_0: all group means are equal, against H_1: at least one mean differs. The F statistic is the ratio of between-group variance to within-group variance. If $p < 0.05$, H_0 is rejected. ANOVA says *that* a difference exists, not *where*.

Effect size (η^2, ω^2). η^2 is the share of total variation explained by block type; ω^2 is a less biased version. Rough guide: 0.01 small, 0.06 medium, 0.14 large. A significant p-value with a large η^2 means the difference is both real and practically important.

Assumption checks. ANOVA assumes normally distributed residuals (Shapiro–Wilk $p > 0.05$) and equal variances (Levene $p > 0.05$). When either is doubtful, Welch ANOVA (unequal variances) and Kruskal–Wallis (rank-based, no normality assumption) are used as confirmations.

Tukey HSD. A post-hoc test that compares every pair of groups while holding the overall error rate at 5%. It produces the mean difference, a confidence interval and an adjusted p-value. If the interval does not contain 0, the pair is significantly different. Letters summarise this: groups that share a letter are not significantly different.

Cohen's d. Size of a pairwise difference in standard-deviation units. 0.2 small, 0.5 medium, 0.8 large.

Two-way ANOVA. Tests two factors at once — block type (mix) and curing age — plus their interaction. A significant interaction means the mixes gain strength at different rates, so the gap between them changes with age.

MANOVA. Tests whether the groups differ on several properties considered together, accounting for the correlations between them. Wilks' λ near 0 means the groups are strongly separated; Pillai's trace is the most robust of the four statistics.

Correlation (r). Measures how strongly two properties move together across specimens (–1 to +1). $|r| > 0.7$ is strong.

Composite index. Converts each property to a 0–1 score among the experimental mixes (1 = best) and averages them with equal weights, giving a single overall ranking.

2. Interpretation of each test

Weight in air. The experimental blocks weigh 4.64–4.66 kg against 14.16 kg for CHB, a reduction of about 67%. The all-group ANOVA is highly significant ($p < 0.0001$), but among Exp 1–4 there is no difference ($F = 0.58$, $p = 0.636$); all four mixes produce blocks of the same weight. CVs of 0.5–0.7% show very uniform production. Note: CHB variances are larger than the Exp variances (Levene $p = 0.0012$) and residuals are non-normal because of the large scale difference; Welch ANOVA ($p < 0.0001$) confirms the conclusion.

Unit weight. Exp blocks average 875–878 kg/m³ versus 1236 kg/m³ for CHB (about 29% lower density), placing them in the lightweight-masonry range. There is no difference among Exp 1–4 ($p = 0.636$). Unit weight is perfectly correlated with weight in air ($r = 1.00$) for the Exp specimens, showing it was calculated from the weight using a constant block volume, so the two tests carry the same information.

Compressive strength – 7 days. All Exp mixes are 5–6 times stronger than CHB (194 psi). Among the Exp mixes the difference is significant ($F = 5.38$, $p = 0.0056$, $\eta^2 = 0.40$). Exp 2 (1179 psi) and Exp 3 (1141 psi) are significantly stronger than Exp 4 (1002 psi); Exp 1 lies in between.

Compressive strength – 14 days. The Exp-only ANOVA is significant ($F = 7.45$, $p = 0.0011$, $\eta^2 = 0.48$). Exp 1 (1515 psi) and Exp 2 (1486 psi) are significantly stronger than Exp 4 (1210 psi). Exp 1 has overtaken Exp 2 and Exp 3, i.e. it gains strength faster after the first week. Shapiro–Wilk $p = 0.050$ is borderline, but Kruskal–Wallis ($p = 0.0072$) agrees.

Compressive strength – 28 days. This is the design-relevant age. The Exp-only effect is the strongest of all tests ($F = 11.16$, $p < 0.0001$, $\eta^2 = 0.58$): 58% of the variation among the Exp blocks is explained by mix. Exp 1 (1806 psi), Exp 2 (1795 psi) and Exp 3 (1695 psi) form one statistically equal group (letter a); Exp 4 (1465 psi) is significantly lower than all three (Cohen's $d > 1.8$). Compared with CHB (457 psi), the Exp mixes are 3.2–4.0 times stronger. In MPa the Exp mixes reach about 10.1–12.5 MPa against 3.2 MPa for CHB.

Strength development (two-way ANOVA). With all groups, mix, age and the mix × age interaction are all significant (interaction $p < 0.0001$): CHB develops strength on a very different curve (only 42% of its 28-day strength at 7 days, versus 60–68% for the Exp mixes). Among Exp 1–4 only, mix ($p < 0.0001$) and age ($p < 0.0001$) are significant but the interaction is not ($p = 0.068$), meaning the four Exp mixes gain strength along roughly parallel curves and their ranking is stable over time — Exp 4 stays lowest at every age. The Exp blocks reach most of their strength early, which is useful for early handling and stacking.

Screw pull-out. Exp mixes hold screws 23.7–27.7 kg against 9.4 kg for CHB (2.5–2.9×). Among Exp mixes the ANOVA is significant ($p = 0.0189$, $\eta^2 = 0.33$), but Tukey finds only one significant pair: Exp 2 > Exp 4. CHB has the largest CV (18%), meaning its fastening capacity is the least predictable.

Split tensile strength. Exp mixes reach 0.341–0.393 MPa versus 0.110 MPa for CHB (3.1–3.6×). There is **no significant difference** among Exp 1–4 ($F = 1.80$, $p = 0.174$; Kruskal–Wallis $p = 0.166$). Exp 4 is numerically lowest, consistent with its compressive results, but the difference is within experimental scatter; Exp 1 and Exp 2 have high CVs (15–16%), which reduces the power to detect differences. The tensile/compressive ratio (≈3.0–3.4%) is similar for all groups, indicating normal brittle behaviour.

Moisture absorption. Exp blocks absorb 8.21–8.66% versus 13.33% for CHB, about 35–38% less water, which indicates a denser or less porous matrix and better durability. Among the Exp mixes the effect is significant ($p = 0.0086$): Exp 3 absorbs significantly less than Exp 1 and Exp 4; Exp 2 is statistically equal to Exp 3. Lower absorption is better.

MANOVA. Considering compressive strength, split tensile, screw pull-out, moisture absorption and unit weight together, the four Exp mixes differ significantly (Wilks' $\lambda = 0.056$, Pillai's trace $p < 0.0001$). The overall multivariate difference is therefore real even though individual properties such as split tensile strength or unit weight do not separate the mixes on their own. With CHB included, the separation is extreme (Wilks' $\lambda \approx 0.0001$).

Correlations. Within the Exp specimens, all strength properties are strongly and positively related (compression 28 d with 14 d $r = 0.95$, with screw pull-out $r = 0.86$, with split tensile $r = 0.81$). A stronger block is therefore also better at holding screws and resisting splitting. Moisture absorption correlates negatively with strength ($r = -0.39$ to -0.61), as expected: more porous blocks absorb more water and are weaker. Block weight is **not** correlated with strength ($r \approx 0$), so within this weight range strength depends on the mix, not on how heavy the block is.

Efficiency and composite ranking. Specific strength (28-day strength per unit weight) is 1.67–2.06 psi per kg/m³ for the Exp mixes versus 0.37 for CHB, i.e. 4.5–5.6 times more structurally efficient. On the equal-weighted composite index the ranking is Exp 2 (0.85), Exp 3 (0.74), Exp 1 (0.52), Exp 4 (0.10). Min–max scaling magnifies small gaps: the underlying differences among Exp 1, Exp 2 and Exp 3 are mostly not statistically significant, whereas Exp 4 is clearly last.

3. Overall conclusions

- All four experimental mixes significantly outperform the conventional CHB in every test: they are about 67% lighter, 3.2–4.0 times stronger in compression at 28 days, 2.5–2.9 times better in screw pull-out, 3.1–3.6 times stronger in split tension and absorb about 35–38% less water.
- The experimental mixes do not differ in weight, unit weight or split tensile strength.
- Exp 4 is consistently the weakest experimental mix in compression at every age and in screw pull-out, and it is statistically separated from the others at 28 days.
- Exp 1, Exp 2 and Exp 3 are statistically equivalent in 28-day compressive strength. Exp 2 has the best overall balance (highest composite index, highest screw pull-out, lowest unit weight, early strength), Exp 1 has the highest 28-day strength and fastest strength gain, and Exp 3 has the lowest moisture absorption and generally the most consistent results (lowest CVs).
- Recommended mix: **Exp 2** on overall performance (composite index 0.85), with Exp 1 an equally valid choice where 28-day compressive strength is the governing criterion and Exp 3 where low water absorption is the priority. Exp 4 is the least suitable.

4. Limitations

- Sample size is 7 per group; small differences (e.g. in split tensile strength) may exist but cannot be detected with this many specimens.
- CHB has a much larger and more variable scale than the Exp blocks in weight in air, producing unequal variances and non-normal residuals in that all-group test; Welch ANOVA was used to confirm the result.

- The two-way ANOVA treats specimens at 7, 14 and 28 days as independent (compression testing is destructive). The correlation analysis assumes replicate i in each sheet refers to the same specimen, following the layout of the Raw Files sheet.
- The composite index uses equal weights; a different weighting (e.g. emphasising compressive strength) can change the order of the closely matched Exp 1, Exp 2 and Exp 3, but not the last place of Exp 4.
- The "Moisture Absorption" sheet has no matching table in the Raw Files sheet; its values were used as given.